Week 7 Software Lesson: Survival Analysis in R

Goals

The goal of this software lesson is to learn how to
- plot Kaplan-Meier curves,
- estimate quartile (e.g., Q1, median, Q3) survival time(s), and
- estimate survival at specific time points, and
- conduct hypotheses tests for comparing two groups on time-to-event data (log-rank test, Gehan-Breslow-Wilcoxon test).

Required Packages

You will need to install and load the following packages in R for this software lesson:
- `survival`
- `ggfortify`

DATASET

In this software lesson, you will use the Laryngoscope dataset. The dataset, data dictionary, and dataset introduction can be found on Canvas, titled *Laryngoscope.csv*, *Laryngoscope Data Dictionary.pdf*, and *Laryngoscope Dataset Introduction.pdf*, respectively.

```
lar <- read.csv(file = file.choose(), header = TRUE)
```

SURVIVAL ANALYSIS

Thus far in the course, you have focused on *linear* regression, which is used when you have a continuous outcome, and *logistic* regression, which is used when you have a categorical outcome. Another type of variable that is common in the health sciences is time-to-event, which occurs when participants are followed for a period of time until a specified event occurs. Survival analysis is the statistical method for analyzing time-to-event data.

Because of the unique nature of the data, two pieces of information are needed for each participant:

- the length of time the participant was followed, and
- whether or not the participant had the event of interest.

These two variables together make up the time-to-event outcome in the analysis.

In the Laryngoscope study, the researchers were interested in comparing a novel laryngoscope (Pentax) with a standard one (Macintosh) on time-to-intubation on the first try. In this example, `Randomization` is the intervention variable (1 denoted for those that received *Pentax* and 0 for those that received *Macintosh*), and the two variables that make up the time-to-event outcome are `attempt1_time` (how many seconds it took to intubate) and `attempt1_S_F` (1 denoted as successful intubation on first try and 0 for not successful intubation (censored)).

Create Kaplan-Meier survival curve

The best way to visualize time-to-event data is via a Kaplan-Meier plot, either overall or by groups. Since examining survival by group is more common, this will be shown first.

- To create Kaplan-Meier survival curves by group,
  - first, use the `survfit()` function from the `{survival}` package to calculate the survival at each time point via the Kaplan-Meier method, specifying the model in the form of `y ~ x`, with `y` being the time-to-event outcome, created using the `Surv()` function with the time variable in the first argument and the event indicator in the second argument, and `x` being the grouping variable, and the name of the dataset (`data =`). Save the information under an object name for later use (e.g., `lar_surv`).
  - Then, use the `autoplot()` function from the `{ggfortify}` package to create the Kaplan-Meier plot, specifying the saved `survfit` object (e.g., `lar_surv_bygrp`) and any additional arguments or functions for customizing the output:
    * `conf.int = FALSE` to suppress confidence bands around the lines;
    * `censor.shape=` to change shape of the censored observations
    * the `xlab()` or `ylab()` functions for labeling the axes;
    * the `labs()` function to label the title, subtitle, caption, and/or legend title;
    * the `theme_bw()` function to modify the theme of the plot; and,
    * `theme()` function to customize text, such as centering the title (`hjust=0.5`) or bolding (`face=bold`)):

```r
## Calculate survival for each time
lar_surv_bygrp <- survfit(Surv(attempt1_time, attempt1_S_F) ~ Randomization, data = lar)

## Create Kaplan-Meier Plot
autoplot(lar_surv_bygrp, conf.int = FALSE, censor.shape = "X") +
  xlab("Time (in seconds)") +
  ylab("Percent Not Succesfully Intubated") +
  labs(title = "Time to Intubation", subtitle = "By Laryngoscope Type",
       caption = "First Attempt Only", color = "Intervention") +
  scale_color_discrete(labels = c("Macintosh", "Pentax")) +
  theme_bw()+
  theme(plot.title = element_text(hjust = 0.5, face = "bold"),
        plot.subtitle = element_text(hjust = 0.5))
```

The plot displays one curve per group. Each `X` on the plot indicates a censored observation (i.e., observations that did not have the event, in this case, an observation with an unsuccessful intubation).

- To create a Kaplan-Meier survival curve of overall survival, use the code you have seen thus far, but change the `x` to be 1:

```r
## Calculate survival for each time
lar_surv <- survfit(Surv(attempt1_time, attempt1_S_F) ~ 1, data = lar)

## Create Kaplan-Meier Plot
autoplot(lar_surv, conf.int = FALSE, censor.shape = "O", censor.size = 4) +
  xlab("Time (in seconds)") +
  ylab("Percent Not Succesfully Intubated") +
  labs(title = "Time to Intubation", subtitle = "Overall",
       caption = "First Attempt Only") +
  theme_bw()+
  theme(plot.title = element_text(hjust = 0.5, face = "bold"),
        plot.subtitle = element_text(hjust = 0.5))
```

The plot displays one curve of overall survival. Each `0` on this plot indicates a censored observation (i.e., observations that did not have the event, in this case, an observation with an unsuccessful intubation).

Compute quantile (e.g., median) survival time(s)

One summary statistic for time-to-event data is median survival time, which is the time when the cumulative survival drops below 50%. The `survfit()` function automatically estimates the median survival time and its 95% confidence interval.

- To obtain median survival time(s) and their CIs, use the `survfit()` function from the `{survival}` package (if you haven't done so already) and then view the saved `survfit` object (e.g., `lar_surv_bygrp`):

```
lar_surv_bygrp
```

The output includes one row per group (if there is a grouping variable) and the following columns:

- the sample size (`n`),
- the number of observations that had an event (`events`),
- the median survival time (`median`), and
- the lower and upper 95% confidence limits for the median survival time (`0.95LCL` and `0.95UCL`).

The median is the 50th percentile, but you can also compute other quartile survival times (e.g., 25th (Q1) or 75th (Q3) percentiles).

- To obtain other quartile survival time(s) and their CIs, use the `quantile()` function from the `{survival}` package, specifying the saved `survfit` object (e.g., `lar_surv_bygrp`):

```
quantile(lar_surv_bygrp)
```

The output includes three different sections for the 25th percentile (Q1, `25` column), 50th percentile (median, `50` column), and 75th percentile (Q3, `75` column) survival times (and one row per group, if there is a grouping variable).

- the quantile survival time value (`$quantile`),
- the lower and upper 95% confidence limits for the quantile survival times (`$lower` and `$upper`).

Compute survival at specific time points

Other summary statistics for time-to-event data include estimated survival at specific time points.

- To obtain survival at specific time points and their CIs, use `summary()` function on the saved `survfit` object, and include the argument `times=`, containing the values of the time points you want estimates of.

```
summary(lar_surv_bygrp, times=c(15, 30, 45, 60, 75, 90))
```

The output includes survival at the specified times, split by group (if there is a grouping variable) in addition to the following columns:

- the specified times (`time`),
- the number of observations that are still risk at that specified time (`n.risk`)
- the number of observations that had an event at that specified time (`n.events`),
- the estimated survival at the specified time (`survival`),
- the standard error for the survival value (`std.err`), and
- the lower and upper 95% confidence limits for the survival value (`lower 95% CI` and `upper 95% CI`).

Some rows for the specified times may be omitted if the survival probability is 0 or there is no data.

Additional Resources

For additional R resources on survival analysis, see:

- Plotting with survival package (R CRAN)
- Survival Analysis in R (Emily Zabor)
- Survival Analysis Basics (STDHA)

Conduct hypothesis test for comparing two groups on time-to-event outcome

Besides descriptive results (as shown thus far in this software lesson), you may also be interested in using inferential results () to compare two groups on a time-to-event outcome. Two common tests that can be used are the log-rank test or the Gehan-Breslow-Wilcoxon test (simply called Wilcoxon test). Both tests have the same null hypothesis (that survival is identical in the two groups), but they differ in how they weight early verses late events, with the log-rank test weighting all time points equally and Wilcoxon test putting more weight on early events compared to later events.

Log-rank test

- To conduct a log-rank test, use the `survdiff()` function from the `{survival}` package, and similar code you have seen thus far, but include the `rho = 0` argument to indicate the type of test to be carried out (`0` for log-rank test):

```
survdiff(Surv(attempt1_time, attempt1_S_F) ~ Randomization, data = lar, rho = 0)
```

The output includes a table with statistics for the test, as well as the Chi-square test statistic (`Chisq=`) and the associated p-value (`p=`).

Wilcoxon test

- To conduct a Wilcoxon test, use the code you have seen thus far, but instead change `rho = 1` to indicate carrying out a Wilcoxon test:

```
survdiff(Surv(attempt1_time, attempt1_S_F) ~ Randomization, data = lar, rho = 1)
```

The output is similar to that from the log-rank test.